MIL-S-83189A(USAF)
22 February 1971
SUPERSEDING
MIL-S-83189 (USAF)
21 October 1968

MILITARY SPECIFICATION

SPRAYING UNIT, CORROSION CONTROL, TRAILER MOUNTED A/M32M-18A

1. SCOPE

1.1 This specification covers a trailer-mounted, corrosion control spraying unit, designated A/M32M-18A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

-300	Sheet and Tape, Reflective, Nonexposed, Lens, Adhesive Backing
-I-735	Isopropyl Alcohol
TT-P-98	Paint, Stencil, Flat

Military

MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-S-3950	Switch, Toggle, General Specification for
MIL-T-5021	Tests, Aircraft and Missile Welding Operators' Qualifications
MIL-R-5520	Reservoirs: Aircraft, Hydraulic, Non-Separated Type
MIL-L-6081	Lubricating Oil, Jet Engine
MIL-C-6529	Corrosion Preventive, Aircraft Engine
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-E-11275	Engines, Gasoline, Industrial Type, General Specification for
MIL-S-12651	Sprayer, Packaging of
MIL-L-23699	Lubricating Oil, Aircraft Turbine Engines, Synthetic Base
MIL-B-26195	Boxes, Wood-Cleated, Skidded Load-Bearing Base

FSC 1730

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STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Demonstration
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing: with Appropriate Test Methods
MIL-STD-1247	Identification of Pipe, Hose, and Tube Lines for Aircraft, Missile, and Space Systems
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28 Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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American Society of Mechanical Engineers

Unfired Pressure Vessel Code

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, New York 10017.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The spraying unit shall consist of the following components:

Item or System Name	See Requirement
Trailer	3.7.1
Compressed air system	3.7.2
Preservative oil and wash solution delivery system	3.7.3
Water delivery system	3.7.4
Control panel	3.7.5.

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials used in the construction of the spraying unit shall be as specified herein. Materials shall not be hammered, filled, nor otherwise treated to conceal defects. Materials not definitely specified shall be of uniform quality and condition and free of seams, cracks, or other defects which may adversely affect the appearance, strength, endurance, and wear resistance of the spraying unit.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

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3.4.2 Protective treatment. When materials are used in the construction of the spraying unit that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack, or scale with age or extremes of climatic and environmental conditions shall not be used.

3.4.3 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage and normal service.

3.4.3.1 Dissimilar metals. Unless protected from the electrolytic action of corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.4 Magnesium-alloy parts. Magnesium-alloy parts shall not be used in the fabrication of the spraying unit.

3.5 Design and construction. The spraying unit shall be so designed and constructed that parts will not work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, installation, and service.

3.5.1 Reliability. The spraying unit shall have a minimum mean-time-between-failures (MTBF) of 300 hours at a confidence factor of 0.90 (6.3.1).

3.5.2 Maintainability. The spraying unit shall be designed and constructed in accordance with the design guidelines of MIL-STD-470, paragraph titled Establish Maintainability Design Criteria. The guidelines shall be applied wherever appropriate to the subject class or type of equipment and wherever they do not conflict with other requirements specified herein. Where practicable, the design and construction of the spraying unit shall permit maintenance and servicing under adverse weather extremes. All maintenance, servicing, and adjustments which may be economically practicable at lower than depot level shall be accomplished using only common handtools and equipment commonly expected in shops doing maintenance on similar equipment.

3.5.2.1 Maintenance downtime (see 6.3.2). The spraying unit shall be designed and constructed to permit maintenance by military service maintenance personnel with a maximum 5-level skill in not more than the downtimes listed below. Not more than two personnel shall be required simultaneously for any task.

3.5.2.1.1 Corrective maintenance downtime. The mean corrective maintenance downtime for field level maintenance shall be not more than 1 hour with a related maximum corrective maintenance downtime of not more than 3 hours.

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3.5.2.1.2 Maximum preventive maintenance downtime. Maximum preventive maintenance downtime, excluding depot maintenance and major overhaul, shall be not more than 1 hour. This downtime shall include the time to service all required items and calibrate all indicating systems on the spraying unit.

3.5.3 Capability. The spraying unit shall be designed for storing and controlling the delivery of preservative oil, wash solution, and water to the compressor section of aircraft installed turbofan, turbojet, and turboshaft engines to prevent compressor corrosion. The unit shall be capable of utilizing the spray nozzles as specified by the procuring activity (see 6.2).

3.5.4 Safety factor. The spraying unit shall be designed for a safety factor of 3 based on the yield strength of the material.

3.5.5 Compactness. Compact, lightweight components shall be used throughout the fabrication of the spraying unit. When possible, high strength-weight ratio materials shall be used to reduce the gross weight of the unit.

3.5.6 Common parts. The spraying unit shall be designed to provide for the maximum practicable interchangeability of hardware and fasteners by using the minimum number of types and sizes of bolts, screws, nuts, washers, and other common hardware.

3.5.7 Plumbing. All lines, valves, and fittings shall be of sufficient
e to permit operation under maximum flow conditions. The pressure drop
ough the system shall not exceed 5 percent under all flow conditions.
All lines, ducts, and tubing shall be identified in accordance with MIL-STD-1247.

3.5.7.1 Rigid lines. The use of rigid lines shall be held to a minimum. When used, such lines shall be of corrosion-resistant steel. Rigid lines, fittings, and components shall withstand 200 percent of the maximum operating pressure without leakage or permanent distortion. Commercial black pipe and fittings shall not be used. Materials (other than thread lubrication and teflon tape) shall not be used for sealing threaded joints.

3.5.7.2 Flexible lines and fittings. Flexible AN or MS hose with AN- or MS-type fittings shall be used. Hose clamps shall not be used for fluid line couplings. All fluid lines, valves, and fittings shall be of a type rated by their manufacturers for the pressure and service to be encountered and shall withstand 200 percent of maximum operating pressure without leakage or permanent distortion.

3.5.8 Foolproofness. Where improper installation of an item could cause malfunctioning of that item or system in which it is installed, it shall be so designed that the item can only be installed in its proper operating position.

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3.5.9 Welders. Welding shall be accomplished by operators certified in accordance with MIL-T-5021, class A.

3.5.10 Lubrication. All points of friction of the spraying unit controls and equipment shall be lubricated in accordance with MIL-STD-838 to insure satisfactory operation.

3.5.10.1 Frequency of lubrication. The spraying unit shall be so designed that it will not require lubrication at intervals of less than 1 month when operated a minimum of 8 hours per day.

3.5.11 Mobility. The spraying unit shall be designed for type II, group C mobility in accordance with MIL-M-8090, except as specified in 3.7.1.

3.5.12 Air transportability. The spraying unit shall be designed to meet the requirements of MIL-A-8421. The unit shall be loaded into an aircraft in the as-ground-transported condition.

3.5.12.1 Tiedown rings. A minimum of four 10,000-pound capacity side tiedown rings conforming to MIL-A-8421 shall be provided on the spraying unit.

3.5.13 Safety. The paragraphs entitled Hazard definition and System safety precedence of MIL-STD-882 shall apply. Other provisions of MIL-STD-882 shall be as specified by the procuring activity (see 6.2).

3.6 Performance. The spraying unit shall be capable of storing and controlling the delivery of water, wash solution, and preservative oil to the compressor section of aircraft installed engines and after being subjected to the following environmental conditions:

3.6.1 Environmental conditions. The spraying unit shall be capable of operating under and satisfactorily resisting the following environmental conditions without the use of an external heating and cooling source:

a. Operating temperatures ranging from -20° to +125°F

b. Storage temperatures ranging from -80° to +160°F

c. Relative humidity up to 100 percent, including conditions wherein condensation takes place in the form of water

d. Salt-fog atmosphere

e. Fungus growth as encountered in tropical climates

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f. Rainfall as encountered in any locale

g. Dust particles as encountered in desert areas.

3.6.2 Operating positions. The spraying unit shall be capable of functioning satisfactorily when its normal operating plane is at a minimum angle of 10° in any direction from horizontal.

3.7 Details of components

3.7.1 Trailer. A trailer shall be provided and shall meet the mobility requirements of 3.5.11, except as follows:

a. Service brakes and safety chains are not required

b. Fenders shall be provided for all four wheels

c. A rear bumper shall be provided. Components which may be damaged by backing or pushing the trailer from the rear shall be protected.

3.7.2 Compressed air system. A compressed air system shall be provided to pressurize the fluid reservoirs for delivery of preservative oil, wash solution, and water to the aircraft installed engine and furnish air for the service air outlet. A flow diagram depicting the compressed air system is shown on figure 1.

3.7.2.1 Air compressor. An air compressor shall be provided and shall be capable of delivering a minimum of 6.8 cfm of free air at a minimum output pressure of 120 psig continuously when supplied air under the environmental operating conditions specified in 3.6.1. The air compressor shall be designed for 1,000 hours continuous duty.

3.7.2.1.1 Gasoline engine. A gasoline engine shall be provided to drive the air compressor. The engine shall meet the requirements for group I engines as specified in MIL-E-11275. A minimum 2-gallon-capacity fuel tank and an automatic rewind rope starter shall be provided on the engine. A two-position, on-off toggle switch in accordance with MIL-S-3950 shall be provided in the ignition circuit. The muffler assembly and intake assembly shall be provided with an Underwriters'-approved-type spark arrestor.

3.7.2.1.1.1 The power drive between the gasoline engine and air compressor shall be equipped with a guard to prevent personnel injury.

3.7.2.2 Compressor relief valve. An adjustable compressor relief valve shall be provided to control the pressure in the air reservoir at any setting from 70 to 110 ±5 psig and shall be sized to relieve the full rated output of the air compressor.

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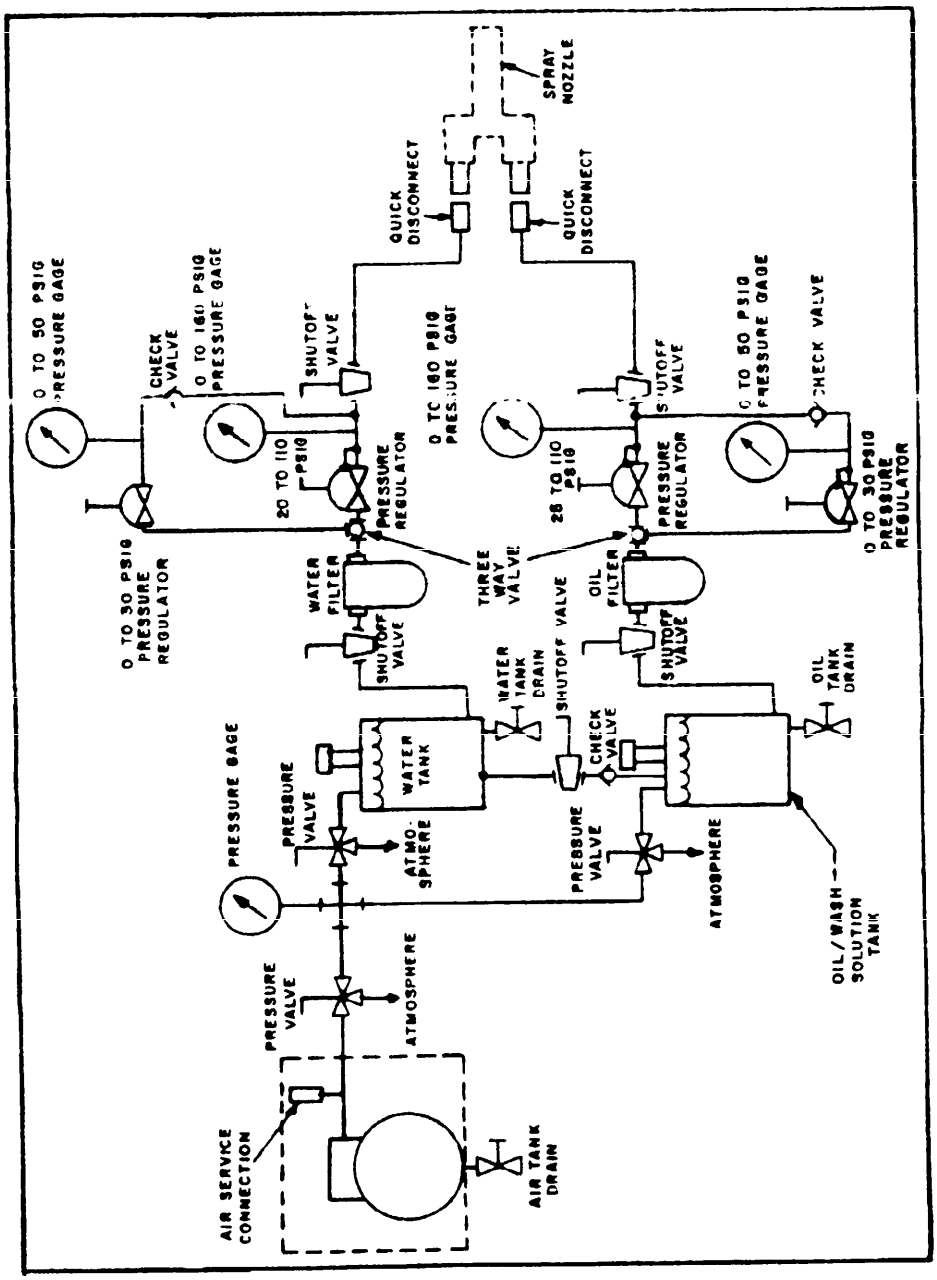


FIGURE 1. Flow Diagram

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3.7.2.3 Air reservoir pressure indicator. A pressure indicator shall be provided and mounted on the air reservoir. The indicator shall have a range and accuracy of 0 to 200 psig and ±5 psig of reading, respectively. The indicator shall be a direct-connected pressure type and shall incorporate a safety pressure relief valve in the pressure line preceding the indicator.

3.7.2.4 Service air connection. A service air connection shall be provided and mounted on the air reservoir outlet. The connection shall be a 1/4-inch, self-sealing, female quick-disconnect.

3.7.2.5 Manual pressure relief valves. A zero leakage (see 6.3.5), manual, pressure relief valve shall be provided on the air reservoir outlet to relieve the air pressure in the air and fluid reservoirs. A zero leakage, manual, pressure relief and shutoff valve shall be provided in each air line upstream of each fluid reservoir to maintain an operating pressure in the preservative oil and wash solution and water delivery systems and to relieve the air pressure in one or both systems as desired by the operator, thereby maintaining air pressure in the air reservoir. The pressure relief and shutoff valves in the air line upstream of each fluid reservoir shall be mounted on the control panel specified in 3.7.5. The air outlets from these valves shall be directed towards the ground.

3.7.2.6 Air reservoir. The air reservoir shall be as specified in 3.7.6.

7.3 Preservative oil and wash solution delivery system. A preservative oil and wash solution delivery system shall be provided to deliver preservative oil and wash solution to the compressor section of aircraft installed engines. The delivery system shall be compatible with and shall not contaminate the preservative oils, wash solution, and water specified in 3.7.3.1 without modifying or changing any component on the spraying unit. The preservative oils and wash solution shall be delivered by the compressed air system specified in 3.7.2. A flow diagram depicting the preservative oil and wash solution delivery system is shown on figure 1.

3.7.3.1 Preservative oils and wash solution. The delivery system shall deliver preservative oils conforming to MIL-L-6081, MIL-C-6529, MIL-L-7808, MIL-L-23699, and Rust-Lick, Incorporated corrosion preventive compound No. 606, or equal, and wash solution consisting of one part Turco Products, Turco 4217, FSN 6850-014-3914, or equal, and 24 parts of demineralized water. The delivery system shall also be compatible with and shall not contaminate demineralized water and fresh water. The delivery system shall be compatible with and shall not contaminate the wash solution, demineralized water, and fresh water when mixed with 50 percent isopropyl alcohol conforming to TT-I-735 or 50 percent methyl alcohol.

3.7.3.2 Preservative oil and wash solution reservoir. The preservative oil and wash solution reservoir shall be as specified in 3.7.6.

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3.7.3.3 Preservative oil and wash solution filter assembly. A preservative oil and wash solution filter assembly shall be provided downstream of the oil reservoir and shall be capable of filtering 10 microns nominal. AN-type filter element(s) shall be utilized. The filter assembly shall be designed for a minimum flow rate of 2 gpm at a pressure of 30 psig.

3.7.3.4 Three-way valve. A manual, zero leakage, three-way valve shall be provided downstream of the filter assembly to direct the fluid through the 0 to 30 or 25 to 110 psig pressure regulators specified in 3.7.3.5. The three-way valve shall be mounted on the control panel specified in 3.7.5.

3.7.3.5 Pressure regulators. Two pressure regulators shall be provided in parallel downstream of the three-way valve specified in 3.7.3.4 for adjusting fluid pressure from 0 to 110 psig. One pressure regulator shall regulate the fluid pressure from 0 to 30 $\pm$0.5 psig and the second pressure regulator shall regulate the fluid pressure from 25 to 110 $\pm$5 psig. The regulators shall be mounted on the control panel specified in 3.7.5. Each regulator shall have a handle for manual pressure adjustment. A positive lock shall be provided for locking the regulator to the desired pressure setting and lock wire provisions shall be provided for permanently locking the regulator to a desired pressure.

3.7.3.6 Pressure indicators. Two pressure indicators shall be provided downstream of the pressure regulators specified in 3.7.3.5 and mounted on the control panel specified in 3.7.5. One indicator shall have a range of 0 to 50 psig with an accuracy of $\pm$0.5 psig of reading and located downstream of the 0 to 30 psig regulator. The other indicator shall have a range of 0 to 160 psig with an accuracy of $\pm$5 psig of reading and located downstream of the 25 to 110 psig regulator. Each indicator shall be a direct-connected pressure type and shall incorporate an automatic resetting safety pressure relief valve in the pressure line preceding the indicator.

3.7.3.7 Check valve. A check valve shall be placed downstream of the 0 to 50 psig pressure gage to prevent overpressurizing the gage when fluid is flowing through the 25 to 110 psig pressure regulator.

3.7.3.8 Shutoff valves. Two manual, zero-leakage, shutoff valves shall be provided in the preservative oil and wash solution line system. One valve shall be located immediately downstream of the reservoir, and the second valve shall be located immediately downstream of the pressure regulator. The second valve shall be mounted on the control console. Each valve shall be of the full-open, full-closed type.

3.7.3.9 Interconnecting preservative oil and wash solution lines. All interconnecting preservative oil and wash solution lines shall have a minimum of 3/8-inch inside diameter.

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3.7.3.9.1 Interconnecting delivery hose assembly. An interconnecting delivery hose assembly conforming to MS28741-6-7800 shall be provided for supplying preservative oil and wash solution to the aircraft installed engine. One end of the hose shall be attached to the spraying unit preservative oil and wash solution system outlet, and the other end shall be provided with the P/N 2DF2 portion of the quick-disconnect of Breco Division, Perfecting Service Company, or interchangeable therewith.

3.7.4 Water delivery system. A water delivery system shall be provided to deliver water (see 6.3.4) to rinse the compressor section of aircraft installed engines. The delivery system shall be compatible with and shall not contaminate the different types of water and water solutions specified in 3.7.4.1 without modifying or changing any component on the spraying unit. The water shall be delivered by the compressed air system specified in 3.7.2. The functional requirements specified for the three-way valve (3.7.3.4), pressure regulators (3.7.3.5), pressure indicators (3.7.3.6), check valve (3.7.3.7)and shutoff valves (3.7.3.8) are also applicable to this system. A flow diagram depicting the water delivery system is shown on figure 1.

3.7.4.1 Water and water solutions. The water delivery system will be utilized with demineralized water, fresh water, and demineralized or fresh water mixed with one or more of the following at one time:

a. Wetting agent Dioctyl Sodium Sulfosuccinate, active ingredient, liquid form, water and mutual solvent, 10 percent concentration (1 quart FSN 6850-[illegible]5-9432)

b. Isopropyl alcohol conforming to TT-I-735; mixed 50 percent water and 50 percent alcohol

c. Methyl alcohol; mixed 50 percent water and 50 percent alcohol.

3.7.4.2 Water reservoir. The water reservoir shall be as specified in 3.7.6.

3.7.4.3 Water filter assembly. A water filter assembly shall be provided downstream of the water reservoir and shall be capable of filtering 10 microns nominal. AN-type filter element(s) shall be utilized. The filter assembly shall be designed for a minimum flow of 25 gpm at a pressure of 30 psig.

3.7.4.4 Supply line. A supply line shall be provided to permit water from the water reservoir to be supplied to the preservative oil and wash solution reservoir. A manual, zero-leakage, shutoff valve shall be provided in the supply line to control the water flow between the reservoirs. The shutoff valve shall be located near the preservative oil and wash solution reservoir to permit the operator to control the valve and observe the reservoir gage at

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the same time. A check valve shall be provided in the supply line to permit only water flow from the water reservoir to the preservative oil and wash solution reservoir.

3.7.4.5 Interconnecting water lines. All interconnecting water lines shall have a minimum of 3/4-inch inside diameter.

3.7.4.5.1 Interconnecting hose assembly. An interconnecting hose assembly conforming to MS28741-12-7800 shall be provided for supplying water to the aircraft installed engine. One end of the hose shall be attached to the spraying unit water system outlet and the other end shall be provided with the P/N 4DF4 portion of the quick-disconnect of Breco Division, Perfecting Service Company, or interchangeable therewith.

3.7.5 Control panel. A control panel shall be provided and mounted on the spraying unit. The control panel shall be a minimum distance of 35 inches from ground level with the panel inclined upwards and away from the operator at a 45° angle from ground level. The control panel may be extended horizontally by a vertical panel next to where the operator will be standing and extending vertically to where the panel starts to become inclined upwards. The maximum reach for the operating controls from the operator's normal position shall not exceed 28 inches. The control panel shall contain the items specified herein.

3.7.5.1 Flow diagram. A basic flow diagram shall be provided on the control panel for the air, preservative oil and wash solution, and water systems.

3.7.5.2 Working surface. A flat, desk-type working surface area, 12 by 14 inches, shall be provided to the right of the control panel. The surface area shall be sloped a maximum of 2° to prevent the accumulation of water on the surface. The working surface need not be connected to the control panel.

3.7.5.3 Timer. A manually operated timer shall be provided and mounted on the control panel. The timer shall have a minimum time duration of 5 minutes. The timer shall be provided with increment markings for each 15 seconds with each increment spacing not less than 1/16 inch, and for each 15 seconds. The necessary controls for the timer shall be mounted on the control panel.

3.7.5.4 Readability. The 0 to 50 psig pressure indicators shall have scales so selected that each division represents 1 psig and that each spacing between the 1-psig spacing is not less than 1/16 inch apart. The remaining pressure indicators shall have scales so selected that each division represents 5 psig and that each spacing between the 5-psig spacings is not less than 1/16 inch apart.

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3.7.5.5 Markings. All control panel markings shall be permanent such as produced by engraving or etching. The markings shall be white on a black background.

3.7.6 Reservoirs. Air, preservative oil and wash solution, and water reservoirs shall be provided. Each reservoir and all components used in the reservoir assembly that will contact the fluid shall be fabricated of stainless steel. Each reservoir shall conform to MIL-R-5520 regarding structural strength, baffles, fluid connections, sump, drain, return line inlet (if required), scuppers, and finish for the basic design requirements and intended use of the spraying unit. Each reservoir shall be designed to withstand an operating pressure of 110 psig and a structural proof test pressure of 240 psig.

3.7.6.1 Access plates. One or more access plates of at least 6-inch diameter in accordance with the ASME Unfired Pressure Vessel Code shall be provided in each reservoir to permit visual inspection and complete access for manual cleaning of the interior of the reservoirs.

3.7.6.2 Capacity. The air reservoir shall have a minimum capacity of 4 cubic feet of free air. The preservative oil and wash solution reservoir shall have a minimum usable capacity of 10 gallons plus a 2-gallon airspace above the fluid level. The water reservoir shall have a minimum usable capacity of 200 gallons of water plus 20 gallons of airspace above the water level. Provisions shall be provided to completely drain each reservoir.

3.7.6.3 Fluid level indicator. Mechanical-type fluid level indicators shall be provided on the preservative oil and wash solution and water reservoirs to indicate the quantity of available fluid in the reservoirs. Sight-glass-type gages will not be acceptable. The preservative oil and wash solution level indicator shall be marked in quarts at the one-fourth, one-half, three-fourths, and full-level positions. The water level indicator shall be marked in gallons at the one-fourth, one-half, three-fourths, and full-level positions. The indicators shall be properly marked to prevent overfilling of the reservoirs and shall be accurate to within ± 2 percent of full scale. The indicators shall be marked by a red line to indicate the minimum operating level of the fluid in the reservoirs. The indicators shall be mounted directly on the reservoirs in an accessible location.

3.7.6.4 Fill cap. The fill cap on the preservative oil and wash solution and water reservoirs shall be either a type that cannot be removed while the reservoir is pressurized or a type that automatically depressurizes while being removed.

3.7.7 Thermal expansion relief. Relief valves shall be installed in the air, preservative oil and wash solution, and water systems wherever necessary to accomplish thermal expansion relief.

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3.7.8 Hose racks. Two hose racks shall be provided to store the preservative oil and wash solution and water interconnecting hoses on the trailer. The mating half of the quick-disconnects specified in 3.7.3.9.1 and 3.7.4.5.1 shall be provided near the hose racks for securing the hose ends securely to the spraying unit.

3.8 Handbook and record compartment. A compartment shall be provided for containing the equipment operation handbook and the equipment maintenance and inspection records. The compartment shall be watertight (see 6.3.3) and shall be a minimum size of 10 inches wide by 12 inches high by 2 inches deep. This compartment may serve as the working surface specified in 3.7.5.2.

3.9 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.10 Electromagnetic interference. The spraying unit shall meet the requirements of MIL-STD-461 and MIL-STD-462 for class D equipment.

3.11 Dimensions. The maximum dimensions of the spraying unit shall be as follows:

a. Length - 96 inches with towbar in vertical position

b. Height - 66 inches with towbar in vertical position

c. Width - 60 inches.

3.12 Weight. The maximum dry weight of the spraying unit as packaged for air shipment shall be 1,600 pounds.

3.13 Threads

3.13.1 Screw threads. Screw threads shall conform to Handbook H28.

3.13.2 Pipe threads. Pipe threads shall conform to MIL-P-7105.

3.14 Finishes and protective coatings. Except as specified, cleaning, painting, plating, and anodic film shall be in accordance with MIL-STD-808 as specified for type I exposure. The exterior surfaces shall be finished in yellow, film designation DG. The control panel background shall be black, film designation UL.

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3.15 Operational markings

3.15.1 Nameplates, identification plates, and placards. All nameplates, identification plates, and placards shall be in accordance with MIL-P-514. All plates shall be attached by metal screws or rivets.

3.15.1.1 Vendor's nameplate. The vendor's standard nameplate may be installed in addition to the nameplate of the procuring activity.

3.15.1.2 Placards. All valves, instruments, controls, outlets, and connections shall be identified by a name-and-number placard. The same numbers shall be shown on the flow and circuit diagrams and other data furnished with the spraying unit.

3.15.1.2.1 Warning. Warning placards or other precautionary labels as necessary shall be so designed and located that the operator cannot fail to notice them before controls, valves, and switches are operated. Warning placards shall be of laminated, phenolic sheet with red-surface backing to result in a red placard with white, engraved letters.

3.15.2 Stenciling

3.15.2.1 The following information shall be stenciled in 1-inch gloss black letters, color 37038 conforming to TT-P-98, on each of the four sides of the spraying unit:

FOR AIR SHIPMENT

THIS UNIT IS DESIGNED FOR

* G FORE AND AFT,

* G SIDEWARD

* To be filled in by the contractor.

3.15.2.2 The following information shall be stenciled in 1-inch gloss black letters, color 37038 conforming to TT-P-98, on each tiedown and lifting point:

TIEDOWN POINT

OR

LIFTING POINT

GROSS WEIGHT *

* To be filled in by the contractor.

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3.15.2.3 The following information shall be in reflectorized adhesive sheeting conforming to L-S-300 and placed on the towbar or front of the spraying unit. The adhesive shall have a red background with 1-inch silver letters and numerals.

MAXIMUM SPEED - 20 MPH.

3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17 Cleaning. The spraying unit shall be thoroughly cleaned. Metal chips and other foreign materials shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the spraying unit shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be conducted at an atmospheric pressure of 29.5 to 30 inches Hg, temperatures of 70° ±5°F, and a relative humidity of 60 to 70 percent. When tests are conducted with atmospheric pressure, temperature, and humidity differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Preliminary run-in and calibration. Preliminary run-in of the spraying unit as may be recommended by the manufacturer shall be performed prior to submission of the unit for testing. A certified log of all preliminary running time, calibration date, et cetera, shall be submitted with the spraying unit.

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4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One spraying unit shall be subjected to the tests specified in 4.4.3.

4.4.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded.

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning.

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections.

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions.

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item.

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests specified under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the individual tests.

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bearing backlash shall be as specified on drawings 2517710 and 2517715.

4.5.1 Individual tests. Each spraying unit shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. System tests . . . . . See 4.6.2

c. Reservoir proof pressure tests . . . . . See 4.6.5.

4.6 Test methods

4.6.1 Examination of product. The spraying unit shall be inspected to determine compliance with the requirements specified herein with respect to parts, tightness of fittings, screws, bolts; and workmanship, finish, materials, marking, and dimensions.

4.6.1.1 Serviceability. The spraying unit shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, operation, and safety. Particular note shall be made with regard to provisions to prevent accumulation of dirt, water, et cetera, which may hinder servicing and operation.

4.6.2 System tests. The compressed air, preservative oil and wash solution, and water systems shall be tested as complete systems. Any failure resulting from these tests shall be cause for rejection.

4.6.2.1 Compressed air system. The complete compressed air system shall be tested to demonstrate its capability to furnish and maintain compressed air to the preservative oil and wash solution and water systems and service air connection at all pressures from 70 to 110 ±5 psig.

4.6.2.1.1 Compressor relief valve. The compressor relief valve shall be tested to verify that the valve will maintain air reservoir pressure at all pressures between 70 and 110 ±5 psig. Air reservoir pressure settings shall be tested at 70, 80, 90, 100, and 110 ±5 psig and each pressure setting shall be maintained for a minimum period of 5 minutes.

4.6.2.2 Preservative oil and wash solution delivery system. With the reservoir pressurized to 110 ±5 psig, the delivery system shall be tested to verify that preservative oil will be delivered through the delivery hose at all oil pressure settings from 5 to 110 psig. Oil pressure settings shall be tested at 5, 10, 15, 25, 30, 50, 70, 90, 100, and 110 psig and each pressure shall be maintained for a minimum period of 5 minutes. Oil shall be delivered through the delivery hose at each of the pressure settings specified for a minimum period of 1 minute.

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4.6.2.3 Water delivery system. With the water reservoir pressurized to 110 +5 psig, the water delivery system shall be tested to verify that fresh water will be delivered through the delivery hose at all water pressure settings from 5 to 110 psig. Water pressure settings shall be tested at 5, 10, 15, 25, 30, 50, 70, 90, 100, and 110 psig and each pressure shall be maintained for a minimum period of 5 minutes. Water shall be delivered through the delivery hose at each of the pressure settings specified for a minimum period of 1 minute.

4.6.2.4 Pressure indicators. Each pressure indicator specified in 3.7.2.3, 3.7.3.6, and 3.7.4 shall be tested for calibration and shall be tested during their respective systems test.

4.6.2.5 Fluid level indicators. The preservative oil and wash solution and water fluid level indicators shall be tested for the accuracy requirements specified in 3.7.6.3.

4.6.3 Air compressor test. The air compressor shall be tested to verify that it will deliver a minimum of 6.8 cfm of free air at a minimum output pressure of 120 psig continuously when supplied air under the environmental operating conditions specified in 3.6.1. During this test, the air compressor shall be driven by the gasoline engine specified in 3.7.2.1.1.

4.6.4 Manual pressure relief and shutoff valves. The relief and shutoff valves specified in 3.7.2.5, 3.7.3.8, and 3.7.4 shall be tested for zero leakage (see 6.3.5) at a minimum operating pressure of 110 psig on the upstream side of the valve and atmospheric pressure on the downstream side of the valve.

4.6.5 Reservoir proof pressure test. The air, preservative oil and wash solution, and water reservoirs shall be subjected to a minimum structural proof test pressure of 240 psig in accordance with MIL-R-5520.

4.6.6 Rigid and flexible fluid lines. Rigid and flexible fluid lines, and components thereof, shall be subjected to 200 percent of maximum operating pressures. There shall be no leakage or permanent distortion as a result of this test.

4.6.7 Fill caps. The fill caps on the preservative oil and wash solution and water reservoirs shall be tested to verify that the caps cannot be removed while the reservoir is pressurized or automatically depressurizes the reservoir while it is being removed.

4.6.8 Preservative oil and wash solution and water filters. The preservative oil and wash solution and water filters shall be tested to verify the minimum flow rate requirements specified in 3.7.3.3 and 3.7.4.3.

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4.6.9 Preservative oil and wash solution compatibility. The preservative oil and wash solution system shall be tested to verify that the system is compatible with the preservative oils, wash solution, and water specified in 3.7.3.1.

4.6.10 Water delivery system. The water delivery system shall be tested to verify that the system will be compatible with the wetting agent and water solutions specified in 3.7.4.1.

4.6.11 Timer. The timer shall be tested to verify compliance with the requirements specified in 3.7.5.3.

4.6.12 Towing tests. The spraying unit shall be tested in accordance with MIL-M-8090 for type II, group C mobility. All components shall remain securely fastened in place and in an operable condition. Upon completion of the mobility tests, the unit shall be subjected to the system tests specified in 4.6.2. Any failure of these tests shall be cause for rejection.

4.6.13 Tiedown and lifting tests. The spraying unit shall be secured during this test by means of the tiedown rings only and each load shall be applied at least once in accordance with MIL-A-8421. Upon completion of all tests, except for the emergency landing load test, the unit shall be serviced and the system tests specified in 4.6.2 shall be conducted. Any failure of these tests shall be cause for rejection.

4.6.14 Electromagnetic interference test. The spraying unit shall be tested to and shall comply with the requirements specified in 3.10.

4.6.15 Operation in tilted position. The spraying unit shall be tilted in eight different directions, 45° apart on a minimum 10° incline with the true horizontal plane, and subjected to the tests specified in 4.6.2.1, 4.6.2.2, 4.6.2.4, and 4.6.2.5.

4.6.16 Environmental tests. The spraying unit shall be subjected to the following tests conducted in accordance with the specified procedures of MIL-STD-810. Upon completion of each test, the unit shall be subjected to the system tests specified in 4.6.2. Failure of the spraying unit to operate and meet the accuracies specified during or after each exposure, failure of any item or assembly due to deterioration from exposures in these tests, or formation of any undesirable condition that cannot be remedied by routine maintenance shall be cause for rejection.

4.6.16.1 Low temperature test. The spraying unit shall be subjected to low temperature storage in accordance with method 502, procedure I. The temperature shall then be raised to -20° ±3°F and stabilized. The unit shall be serviced

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with water mixed as specified in 3.7.4.1 and oil at a temperature of -20° ±3°F. The system tests specified in 4.6.2 shall then be conducted. The temperature shall then be raised to 32° ±3°F and stabilized. The water system shall be reserviced with 100 percent fresh or demineralized water at 32° ±3°F and then the system tests specified in 4.6.2 shall be conducted. All indicators, valves, et cetera, shall perform satisfactorily and meet the specified requirements at the -20° and +32°F conditions.

4.6.16.2 High temperature test. The spraying unit completely serviced shall be subjected to high temperature storage in accordance with method 501, procedure I. The temperature shall then be lowered to 125° ±3°F and stabilized. The system tests specified in 4.6.2 shall then be conducted. All indicators, valves, et cetera, shall perform satisfactorily and meet the specified requirements in the 125°F condition.

4.6.16.3 Humidity test. The spraying unit shall be subjected to humidity in accordance with method 507, procedure I.

4.6.16.4 Salt-fog test. The spraying unit shall be subjected to salt-fog in accordance with method 509, procedure I for 50 hours.

4.6.16.5 Fungus resistance test. The spraying unit or representative samples of the material or component shall be subjected to fungus in accordance with method 508, procedure I.

4.6.16.6 Rain test. The spraying unit shall be subjected to rain in accordance with method 506, procedure I.

4.6.16.7 Dust test. The spraying unit shall be subjected to dust in accordance with method 510, procedure I.

4.6.17 Reliability demonstration and test. Satisfactory completion of all tests specified herein demonstrates compliance with quantitative reliability requirements of this specification.

4.6.18 Maintainability demonstration. Compliance with the maintainability requirements shall be demonstrated by task performances as specified below. The active maintenance downtimes, the tools and equipment required, and any obstructions or conditions hindering the task of preventing potential hazards to personnel or to parts shall be observed and recorded. Signs of excessive wear or other indications of potential failure shall be recorded where observed and the information supplied to the design activity for study and recommendation of any appropriate corrections to be reflected in future items delivered. Such recommendations shall be subject to procuring activity approval before being implemented or successful repeat of other tests specified herein

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may be required where the test result could be influenced by the corrections. It shall be demonstrated that the spraying unit is in satisfactory operating condition after the maintenance tasks have been performed. Maintenance tasks that may be practicable at depot level and that are not recommended below depot level shall be identified with a prediction of the downtime required for each task and the tools and equipment required. Such depot tasks need not be physically done. The predictions may be based on engineering judgment. comparison to similar task times, and study of the equipment and drawings.

a. Corrective maintenance - Compliance with quantitative corrective maintenance downtime requirements shall be demonstrated by a maintenance demonstration procedure, maintenance task selection, and maintenance task performances in accordance with MIL-STD-471 including appendix A and appendix B, method 1, test plan pair (A1 plus B1)

b. Preventive maintenance - It shall be demonstrated that the maximum preventive maintenance downtime, excluding depot maintenance and major overhaul, is not more than 3 hours. This demonstration shall be conducted by performing all preventive maintenance actions required on the unit. This shall include all instrument systems calibrated, all fluid filters changed, service all required items, et cetera.

4.6.19 Operational test (see 6.2). After all other tests specified herein have been successfully performed, the spraying unit shall be tested utilizing the spray nozzles specified in 3.5.3 on their respective type aircraft. The inability of the unit to satisfactorily deliver preservative oil and wash solution, and water to the aircraft installed engine as required by the specific operating instructions shall be cause for rejection.

4.7 Examination of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine compliance with section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The spraying unit shall be preserved and packaged in accordance with MIL-S-12651.

5.1.2 Level C. The spraying unit shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Each spraying unit, preserved and packaged as specified in 5.1, shall be packed in a container conforming to MIL-B-26195, type II, style A or B, class 1 or 2.

5.2.2 Level B. Each spraying unit, preserved and packaged as specified in 5.1, shall be packed in a container conforming to MIL-B-26195, type I, style A or B, class 1 or 2.

5.2.3 Level C. The spraying unit shall be packed in a manner which affords adequate protection against damage during direct shipment from the supply source to the first receiving activity. This level shall conform to the applicable carrier rules and regulations and may be the suppliers commercial practice, provided the latter meets the requirements of this level.

5.3 Physical protection. Shipping container contents shall be cushioned, anchored, blocked, and braced in accordance with MIL-STD-1186.

5.4 Marking. In addition to any special marking required in the contract or order (see 6.2), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

SPRAYING UNIT, CORROSION CONTROL, TRAILER MOUNTED A/M32M-18A.

6. NOTES

6.1 Intended use. The A/M32M-18A spraying unit is intended to store and control the delivery of water and preservative oil and wash solution to the compressor section of aircraft installed turbofan, turbojet, and turboshaft engines to prevent compressor corrosion.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. The spray nozzles with which the trailer must be compatible (see 3.5.3)

c. Location and conditions for preproduction testing (see 4.4)

d. Locations and conditions for the operational test (see 4.6.19) including:

(1) The number and types of aircraft installed engines to be used

(2) Extent of contractor support and participation during the operational test.

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e. Other provisions of MIL-STD-882 if applicable (see 3.5.13)

f. Applicable levels of preservation, packaging, and packing required (see 5.1 and 5.2)

g. Special shipment marking (see 5.4).

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Mean-time-between-failures. The mean life or mean-time-between-failures is defined as the arithmetical mean (average) of the operating time between failures.

6.3.2 Downtime. Downtime is defined as the active time required to perform the necessary maintenance.

6.3.3 Watertight. Watertight is defined as so constructed that moisture will not enter the compartment.

6.3.4 Fresh water. Fresh water is defined as water obtained from the local water department.

6.3.5 Zero leakage. Zero leakage for the manual pressure relief and shutoff valves is defined as a leakage not to exceed 4 drops per 24 hours with the preservative oil and wash solution and water systems pressurized to 110 $\pm$5 psig. The manual pressure relief valves utilized in the air pressure system will be tested at 110 $\pm$5 psig utilizing water as the testing fluid in lieu of air.

6.4 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 1730-F140

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC-WPAFB-OCT 67 2M

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